

1. 예제 파일 복사.

/DATA/home/edu000/coverage_example.tar를 본인 directory로 복사.

tar -xvf coverage_example.tar 명령어를 사용해 압축 해제.

cd coverage_example

2. 아래 명령어를 사용해 예제 system verilog code를 compile/elaboration 진행.

```
vcs -full64 -sverilog -cm line+cond -cm_dir ./cov/c432 -cm_name BUILD -o simv  
c432bRTL.v tb-c432bRTL.v -cm_cond full
```

option list:

- full64 : 64비트 빌드.
- sverilog : SystemVerilog 문법 허용.
- cm line+cond : 라인 + 조건문(Condition) 커버리지 계측 컴파일.
- cm_dir ./cov/c432 : 커버리지 DB(vdb) 저장 위치(결과: ./cov/c432.vdb).
- cm_name BUILD : 현재 실행 러닝에 붙일 태그(리포트에서 러닝별 구분).
- o simv : 실행파일 이름.
- cm_cond full : 조건 커버리지 범위를 기본보다 넓게(논리/관계 연산의 부분식 포함).

2-1. vcs 동작 시 발생하는 warning의 원인을 파악하고 문제를 수정하라.

```
Warning-[TFIPC] Too few instance port connections  
tb-c432bRTL.v, 16  
stimulus, "c432 C432(E, A, B, C, PA, PB, PC, Chan);"  
The above instance has fewer port connections than the module definition.  
Please use '+lint=TFIPC-L' to print out detailed information of unconnected  
ports.  
  
Warning-[PCWM-W] Port connection width mismatch  
tb-c432bRTL.v, 16  
"c432 C432(E, A, B, C, PA, PB, PC, Chan, );"  
The following 4-bit expression is connected to 1-bit port "PCB" of module  
"c432", instance "C432".  
Expression: Chan  
Instantiated module defined at: "c432bRTL.v", 7  
Use +lint=PCWM for more details.
```

3. 문제 2번 명령어의 결과물로 생성된 실행파일 run.

```
./simv -cm line+cond -cm_dir ./cov/c432 -cm_name smoke
```

option list:

- cm line+cond : 시뮬레이션 시 커버리지 수집을 명시.
- cm_dir ./cov/c432 : 컴파일 때와 동일한 vdb 폴더에 누적.
- cm_name smoke : 현재 실행 러닝의 이름(테스트별 구분용).

```
VCS Coverage Metrics Release V-2023.12-SP2-8_Full64 Copyright (c) 1991-2023 by Synopsys Inc.
15:: 11111111 01111111 01111111 01111111 111 1000
35:: 11111111 10111111 10111111 10111111 111 0111
55:: 11111111 11011111 11011111 11011111 111 0110
75:: 11111111 11101111 11101111 11101111 111 0101
95:: 11111111 11110111 11110111 11110111 111 0100
115:: 11111111 11111011 11111011 11111011 111 0011
135:: 11111111 11111101 11111101 11111101 111 0010
155:: 11111111 11111110 11111110 11111110 111 0001
175:: 11111111 11111111 11111111 11111111 111 0000
195:: 11111111 11111111 01001111 10111111 010 1111
215:: 11111111 11111111 10111111 01000000 010 0111
235:: 11111111 11111111 11011010 11111111 010 1111
255:: 11111111 11111111 11101011 11111111 010 1111
275:: 11111111 11111111 11110011 11111111 010 1111
295:: 11111111 11111111 11111011 11111111 010 0011
315:: 11111111 11111111 11111101 11111111 010 0010
335:: 11111111 11111111 11111110 11111111 010 0001
355:: 11111111 11111111 11111111 11111111 010 0000
375:: 11111111 11111111 11111111 01111111 001 1000
395:: 11111111 11111111 11111111 10111111 001 0111
415:: 11111111 11111111 11111111 11011111 001 0110
435:: 11111111 11111111 11111111 11101111 001 0101
455:: 11111111 11111111 11111111 11110111 001 0100
475:: 11111111 11111111 11111111 11111011 001 0011
495:: 11111111 11111111 11111111 11111101 001 0010
515:: 10000000 10000000 10000000 10000000 001 0001
535:: 01111101 01111101 01111101 01111100 001 0000
555:: 11111111 11101101 00010010 00010010 100 1111
575:: 11111111 11110100 11111111 11111111 100 1111
595:: 11111111 01111111 10000000 10000000 100 1000
615:: 11111111 00111111 11111111 00111111 101 1111
635:: 00000000 00000000 00000000 00000000 000 1111
$finish called from file "tb-c432bRTL.v", line 40.
$finish at simulation time 640

-----
VCS Coverage Metrics: during simulation line, cond was monitored
-----

Coverage status: End of All Coverages ...

VCS Simulation Report
Time: 640 ns
CPU Time: 0.160 seconds; Data structure size: 0.1Mb
Mon Aug 25 15:06:55 2025
```

4. simulation 결과 파일(Line+Condition Coverage)을 merge하여 Report 생성.

```
urg -dir ./cov/c432 -format both -report ./cov/REPORT_c432
```

option list:

- dir ./cov/c432 : 리포팅 vdb 디렉토리 지정.
- format both : HTML + 텍스트 리포트 동시 생성.
- report ./cov/REPORT_c432 : 리포트 출력 경로.

```
[root@SGP3 coverage_example]# urg -dir ./cov/c432 -format both -report ./cov/REPORT_c432
Note: Bumping stack_limit from 8192 to unlimited Kbytes.
URG Version V-2023.12-SP2-8 Copyright (c) 1991 - Synopsys, Inc.
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Note-[URG-RDG] Report directory generated
Report written to directory ./cov/REPORT_c432
```

4-1. cov/REPORT_c432/modinfo.txt에서 조건문의 값마다 coverage 확인.

```
-----
Cond Coverage for Module : c432

Conditions      Total    Covered    Percent
Logical         6         6      100.00
Non-Logical     612      446      72.88
Event           16         16      100.00

LINE          20
EXPRESSION ((&EAb) == 0)
-----1-----

-1- Status
0 Covered
1 Covered

LINE          35
EXPRESSION ((&XEB) == 0)
-----1-----

-1- Status
0 Covered
1 Covered

LINE          50
EXPRESSION ((&XEC) == 0)
-----1-----

-1- Status
0 Covered
1 Covered

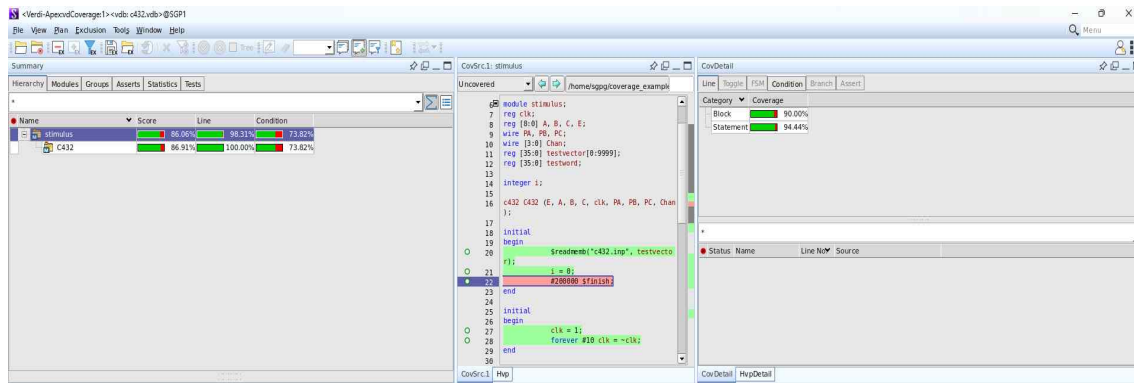
LINE          19
SUB-EXPRESSION BIT 8 of ( ~ (( ~ A ) & E ) )
-----1-----

-1- Status
0 Covered
1 Covered
```

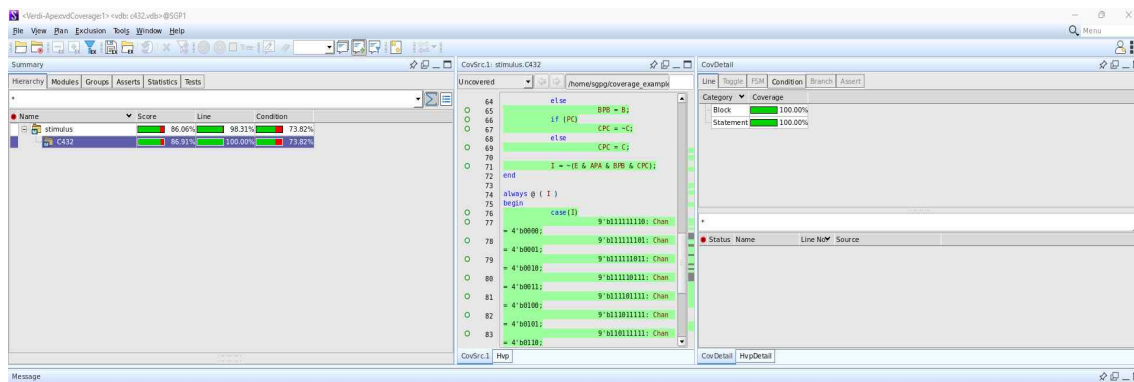
5. Verdi에서 Main Code와 Testbench의 Line/Condition Coverage 분석.

verdi -cov -covdir ./cov/c432.vdb

5-1. Testbench Line Coverage



5-2. Main Code Line Coverage.



5-3. Main Code Condition Coverage.

